20260922 Quiz

for (MATH301) Modern Algebra I

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Exercise 1.1.21. Let G be a finite group and let x be an element of G of order n. Prove that if n is odd, then

$$x = (x^2)^k$$

for some k.

Solution. Since n is odd, $k = (n+1)/2$ is an integer. As $|x| = n$,

$$(x^2)^k = x^{n+1} = x^n x = x.$$

□

Exercise 1.1.25. Prove that if $x^2 = 1$ for all $x \in G$, then G is abelian.

Solution. For every $x \in G$, the hypothesis gives $x^{-1} = x$. Hence for arbitrary $a, b \in G$,

$$ab = (ab)^{-1} = b^{-1}a^{-1} = ba.$$

Thus G is abelian.

□

Exercise 1.1.31. Prove that any finite group G of even order contains an element of order 2.
Hint. Let

$$t(G) = \{g \in G \mid g \neq g^{-1}\}.$$

Show that $t(G)$ has an even number of elements and every nonidentity element of $G \setminus t(G)$ has order 2.

Solution. If $g \in t(G)$, then $g^{-1} \in t(G)$ and $g \neq g^{-1}$. Thus the elements of $t(G)$ occur in disjoint pairs $\{g, g^{-1}\}$, so $|t(G)|$ is even. If $x \in G \setminus t(G)$, then $x = x^{-1}$; hence every nonidentity element of $G \setminus t(G)$ satisfies $x^2 = 1$ and has order 2. Since $|G|$ and $|t(G)|$ are even, $|G \setminus t(G)|$ is even. As $1 \in G \setminus t(G)$, this set contains at least one nonidentity element, and therefore G contains an element of order 2.

□

Exercise 1.1.34. If x is an element of infinite order in G, prove that the elements x^n, $n \in \mathbb{Z}$, are all distinct.

Solution. Suppose $x^m = x^n$ for some $m, n \in \mathbb{Z}$. If $m \neq n$, assume without loss of generality that $m > n$. Then

$$x^{m-n} = 1,$$

with $m - n > 0$, contradicting that x has infinite order. Hence $m = n$, so the powers x^n are all distinct.

□

Exercise 1.2.3. Use the presentation

$$D_n = \langle r, s \mid r^n = s^2 = 1,\ rs = sr^{-1} \rangle$$

to show that every element of D_n which is not a power of r has order 2. Deduce that D_n is generated by the two elements s and sr, both of which have order 2.

Notation. Following MATH000, D_n denotes the dihedral group of symmetries of a regular n-gon, so $|D_n| = 2n$. Dummit–Foote denotes the same group by D_{2n}.

Solution. Every element of D_n which is not a power of r has the form sr^i. From $rs = sr^{-1}$ we obtain $r^i s = sr^{-i}$, so

$$(sr^i)^2 = sr^i sr^i = s(sr^{-i})r^i = 1.$$

Since $sr^i \neq 1$, it has order 2. In particular, s and sr have order 2. Also

$$s(sr) = r,$$

so s and sr generate both r and s, hence generate D_n.

□

Exercise 1.2.6. Let x and y be elements of order 2 in any group G. Prove that if $t = xy$, then

$$tx = xt^{-1}.$$

Thus, if $n = |xy| < \infty$, then x and t satisfy the same relations in G as s and r do in D_n.

Solution. Since $x^{-1} = x$ and $y^{-1} = y$,

$$t^{-1} = (xy)^{-1} = yx.$$

Hence

$$tx = (xy)x = xyx = x(yx) = xt^{-1}.$$

If $n = |t| < \infty$, then

$$t^n = x^2 = 1, \qquad tx = xt^{-1},$$

which are the defining relations for r and s in D_n under $r \leftrightarrow t$ and $s \leftrightarrow x$.

□

Exercise 1.2.7. Show that

$$\langle a, b \mid a^2 = b^2 = (ab)^n = 1 \rangle$$

gives a presentation for D_n in terms of the two generators $a = s$ and $b = sr$ of order 2.

Hint. Show that the relations for r and s follow from the relations for a and b, and conversely, that the relations for a and b follow from those for r and s.

Solution. Set $a = s$ and $b = sr$. Then

$$a^2 = s^2 = 1, \qquad b^2 = (sr)^2 = 1, \qquad ab = s(sr) = r,$$

so $(ab)^n = r^n = 1$.

Conversely, assume $a^2 = b^2 = (ab)^n = 1$ and set

$$s = a, \qquad r = ab.$$

Then $s^2 = 1$ and $r^n = 1$. Since $a^{-1} = a$ and $b^{-1} = b$,

$$r^{-1} = (ab)^{-1} = ba,$$

and therefore

$$rs = (ab)a = aba = a(ba) = sr^{-1}.$$

Thus the two sets of generators and relations determine one another, giving the stated presentation of D_n.

□

Exercise 1.2.15. Find a set of generators and relations for $\mathbb{Z}/n\mathbb{Z}$.

Solution. A generating set is $\{[1]_n\}$, and the defining relation is

$$n[1]_n = [0]_n.$$

Equivalently, in multiplicative presentation notation,

$$\mathbb{Z}/n\mathbb{Z} \cong \langle x \mid x^n = 1 \rangle.$$

□

Exercise 1.3.5. Find the order of

$$(1\ 12\ 8\ 10\ 4)(2\ 13)(5\ 11\ 7)(6\ 9).$$

Solution. The displayed cycles are disjoint and have lengths $5, 2, 3, 2$. Hence the order is

$$\operatorname{lcm}(5, 2, 3, 2) = 30.$$

□

Exercise 1.3.13. Show that an element has order 2 in S_n if and only if its cycle decomposition is a product of commuting 2-cycles.

Solution. Let $\sigma \in S_n$ have order 2. Then $\sigma^2 = 1$. If a nontrivial cycle in the disjoint cycle decomposition of σ had length $m \geq 3$, applying σ twice would not fix the elements in that cycle. Hence every nontrivial cycle has length 2. Since disjoint cycles commute, σ is a product of commuting 2-cycles.

Conversely, if σ is a nonempty product of disjoint 2-cycles, then the cycles commute and each has square 1, so $\sigma^2 = 1$. Since $\sigma \neq 1$, it follows that $|\sigma| = 2$.

□

Exercise 1.3.14. Let p be a prime. Show that an element has order p in S_n if and only if its cycle decomposition is a product of commuting p-cycles. Show by an explicit example that this need not be the case if p is not prime.

Solution. Suppose $|\sigma| = p$. Then $\sigma^p = 1$. If a nontrivial cycle in the disjoint cycle decomposition of σ has length m, then an m-cycle returns to the identity after p applications only when $m \mid p$. Since p is prime and $m > 1$, we have $m = p$. Thus σ is a product of disjoint, hence commuting, p-cycles.

Conversely, a nonempty product of disjoint p-cycles has pth power equal to 1, while no positive power smaller than p is the identity. Hence its order is p.

The primality assumption is necessary. For example,

$$\sigma = (1\ 2)(3\ 4\ 5\ 6)$$

has order 4, but its cycle decomposition is not a product solely of 4-cycles.

□

Exercise 1.3.15. Prove that the order of an element in S_n equals the least common multiple of the lengths of the cycles in its cycle decomposition.

Hint. Use Exercise 1.3.10 and Exercise 1.1.24.

Solution. Write the disjoint cycle decomposition as

$$\sigma = \sigma_1\sigma_2 \cdots \sigma_r,$$

where σ_i has length m_i. Since the cycles are disjoint, they commute, and therefore

$$\sigma^k = \sigma_1^k\sigma_2^k \cdots \sigma_r^k.$$

This is the identity if and only if each σ_i^k is the identity, which holds if and only if $m_i \mid k$ for every i. Hence the least positive such k is

$$\operatorname{lcm}(m_1, \ldots, m_r).$$

□

Exercise 1.3.18. Find all numbers n such that S_5 contains an element of order n.

Hint. Use Exercise 1.3.15.

Solution. The possible nontrivial disjoint cycle types in S_5 have lengths

$$2, \quad 2+2, \quad 3, \quad 3+2, \quad 4, \quad 5.$$

By Exercise 15 their orders are respectively

$$2, \quad 2, \quad 3, \quad 6, \quad 4, \quad 5.$$

Including the identity, the possible orders are therefore

$$\boxed{1, 2, 3, 4, 5, 6}.$$

□

Exercise 1.6.4. Prove that the multiplicative groups $\mathbb{R} \setminus \{0\}$ and $\mathbb{C} \setminus \{0\}$ are not isomorphic.

Solution. The element $i \in \mathbb{C} \setminus \{0\}$ has order 4. If the two groups were isomorphic, $\mathbb{R} \setminus \{0\}$ would also contain an element of order 4, since isomorphisms preserve element orders. But if $x \in \mathbb{R} \setminus \{0\}$ satisfies $x^4 = 1$, then $x = \pm 1$, so x has order 1 or 2. This is a contradiction.

□

Exercise 1.6.14. Let G and H be groups and let $\varphi\colon G \to H$ be a homomorphism. Define the *kernel* of φ to be

$$\ker \varphi = \{g \in G \mid \varphi(g) = 1_H\}.$$

Prove that $\ker \varphi$ is a subgroup of G (cf. Exercise 1.1.26). Prove that φ is injective if and only if $\ker \varphi$ is the identity subgroup of G.

Solution. Since $\varphi(1_G) = 1_H$, we have $1_G \in \ker \varphi$. If $a, b \in \ker \varphi$, then

$$\varphi(ab) = \varphi(a)\varphi(b) = 1_H,$$

and

$$\varphi(a^{-1}) = \varphi(a)^{-1} = 1_H.$$

Thus $\ker \varphi$ is closed under products and inverses, so $\ker \varphi \leq G$.

If φ is injective and $g \in \ker \varphi$, then

$$\varphi(g) = 1_H = \varphi(1_G),$$

so $g = 1_G$. Hence $\ker \varphi = \{1_G\}$. Conversely, suppose $\ker \varphi = \{1_G\}$ and $\varphi(a) = \varphi(b)$. Then

$$\varphi(ab^{-1}) = \varphi(a)\varphi(b)^{-1} = 1_H,$$

so $ab^{-1} = 1_G$, and therefore $a = b$. Hence φ is injective.

□

Exercise 1.6.19. Let

$$G = \{z \in \mathbb{C} \mid z^n = 1 \text{ for some } n \in \mathbb{N}\}.$$

Prove that for any fixed integer $k > 1$, the map from G to itself defined by

$$z \mapsto z^k$$

is a surjective homomorphism but is not an isomorphism.

Solution. Define $\varphi(z) = z^k$. Since multiplication in $\mathbb{C}$ is commutative,

$$\varphi(zw) = (zw)^k = z^k w^k = \varphi(z)\varphi(w),$$

so φ is a homomorphism.

Let $w \in G$. Choose $z \in \mathbb{C}$ with $z^k = w$. If $w^n = 1$, then

$$z^{kn} = (z^k)^n = w^n = 1,$$

so $z \in G$ and $\varphi(z) = w$. Thus φ is surjective.

Finally, let $\zeta = e^{2\pi i/k}$. Then $\zeta \neq 1$, $\zeta \in G$, and

$$\varphi(\zeta) = \zeta^k = 1 = \varphi(1).$$

Hence φ is not injective and therefore is not an isomorphism.

□

Exercise 1.6.22. Let A be an abelian group and fix some $k \in \mathbb{Z}$. Prove that the map

$$a \mapsto a^k$$

is a homomorphism from A to itself. If $k = -1$, prove that this homomorphism is an isomorphism, i.e., an automorphism of A.

Solution. For $a, b \in A$, the elements commute, so the exponent laws give

$$(ab)^k = a^k b^k.$$

Hence the map $\varphi(a) = a^k$ satisfies

$$\varphi(ab) = \varphi(a)\varphi(b),$$

and is a homomorphism.

If $k = -1$, then $\varphi(a) = a^{-1}$ and

$$\varphi(\varphi(a)) = (a^{-1})^{-1} = a.$$

Thus $\varphi^{-1} = \varphi$, so φ is an automorphism of A.

□

Exercise 1.6.25. Let $n \in \mathbb{N}$, let r and s be the usual generators of D_n, and let $\theta = 2\pi/n$.

(a) Prove that

$$\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$$

is the matrix of the linear transformation which rotates the xy-plane about the origin in a counterclockwise direction by θ radians.

(b) Prove that the map $\varphi\colon D_n \to \mathrm{GL}_2(\mathbb{R})$ defined on generators by

$$\varphi(r) = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}, \qquad \varphi(s) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

extends to a homomorphism of D_n into $\mathrm{GL}_2(\mathbb{R})$.

(c) Prove that the homomorphism φ in part (b) is injective.

Solution. Denote the matrix in part (a) by $\mathbf{R}$ and the matrix assigned to s in part (b) by $\mathbf{S}$.

(a) A counterclockwise rotation by θ sends

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} \mapsto \begin{pmatrix} \cos\theta \\ \sin\theta \end{pmatrix}, \qquad \begin{pmatrix} 0 \\ 1 \end{pmatrix} \mapsto \begin{pmatrix} -\sin\theta \\ \cos\theta \end{pmatrix}.$$

These images are the columns of $\mathbf{R}$, proving part (a).

(b) Since $\mathbf{R}$ is rotation by $2\pi/n$, $\mathbf{R}^n = \mathbf{I}$, and directly $\mathbf{S}^2 = \mathbf{I}$. Moreover,

$$\mathbf{RS} = \begin{pmatrix} -\sin\theta & \cos\theta \\ \cos\theta & \sin\theta \end{pmatrix} = \mathbf{SR}^{-1}.$$

Thus $\mathbf{R}$ and $\mathbf{S}$ satisfy $r^n = s^2 = 1$ and $rs = sr^{-1}$. Hence $r \mapsto \mathbf{R}$, $s \mapsto \mathbf{S}$ extends to a homomorphism $\varphi\colon D_n \to \mathrm{GL}_2(\mathbb{R})$.

(c) By Exercise 1.6.14, it suffices to show $\ker\varphi = \{1\}$. Every element of D_n is r^i or sr^i with $0 \le i < n$. If $\mathbf{R}^i = \mathbf{I}$, then rotation by $2\pi i/n$ is the identity, so $i = 0$. Also

$$\det(\mathbf{SR}^i) = \det(\mathbf{S})\det(\mathbf{R}^i) = -1 \neq \det(\mathbf{I}),$$

so $\mathbf{SR}^i \neq \mathbf{I}$ for every i. Hence $\ker\varphi = \{1\}$, and φ is injective.

□